

The Nagata embedding theorem

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Summary. Except for a few corrections added in 2010, the article reproduces notes sent to Nagata in the 1970s, which were a translation into the language of schemes of his papers [2] and [3]. The proofs are a “constructive” version of Nagata’s proofs. Making the proofs constructive made it possible to remove the use of valuations. A more detailed exposition and additional results are in Conrad [1]. Unless expressly stated otherwise, all schemes considered are assumed Noetherian.

Abstract. Except for a few corrections added in 2010, the article reproduces notes, sent to Nagata in the 1970s, which were a translation in the language of schemes of his articles [2], [3]. The proofs are a “constructive” version of his proofs. Making the proofs constructive allowed us to proceed without using valuations. A more detailed exposition and additional results are in Conrad [1]. Unless otherwise mentioned, all the schemes considered are supposed to be Noetherian.

0. Reminders on blow-ups

0.1. If $\mathfrak{a}$ is an ideal of the ring A , the scheme obtained from $X = \text{Spec}(A)$ by blowing up $\mathfrak{a}$ is the union of the open sets $D(x)$, where x runs through a system of generators of $\mathfrak{a}$, and $D(x)$ is the spectrum of the subring $A[\mathfrak{a}x^{-1}]$ of $A[x^{-1}]$. Moreover,

$$D(x) \cap (X - V(\mathfrak{a})) = \text{Spec}(A[x^{-1}]).$$

If $\mathfrak{c}$ is a second ideal of A , defining a subscheme $V(\mathfrak{c})$ of X , the trace on $D(x)$ of the proper transform of $V(\mathfrak{c})$ is defined by the ideal of $A[\mathfrak{a}x^{-1}]$ obtained as the trace of the ideal of $A[x^{-1}]$ generated by $\mathfrak{c}$.

Lemma (0.2). *Let X be a scheme, let $\mathfrak{a}$ and $\mathfrak{b}$ be two sheaves of ideals on X , and let $\tilde{X}$ be the blow-up of X along $\mathfrak{a} + \mathfrak{b}$. Then the open subset of $\tilde{X}$ which is the union of the $D(x)$, for $x \in \mathfrak{b}$, contains the proper transform of $V(\mathfrak{a})$.*

Indeed, $\tilde{X}$ is the union of the $D(x)$, for $x \in \mathfrak{a}$ or $x \in \mathfrak{b}$, and, for $x \in \mathfrak{a}$, $D(x)$ is disjoint from the proper transform of $V(\mathfrak{a})$.

Lemma 0.3 of the original text was false. The corrected version below dates from March 1996. It brought about a change of notation. The original lemma used the following notation, to which the rest of the text refers: rather than F and G , one considered their complements U and V , which, with $Y' := U \cap p^{-1}(Y)$, formed a commutative diagram

$$\begin{array}{ccc} Y' & \longrightarrow & Z \\ \downarrow & & \downarrow p \\ U & \hookrightarrow & V \hookrightarrow X. \end{array}$$

where Y' is closed in $p^{-1}(V)$.

Lemma (0.3). *Let $p : Z \rightarrow X$, let $F \supset G$ be closed subschemes of X , and let Y be a subscheme of Z . Assume that*

$$Y \cap p^{-1}(F) \subset p^{-1}(G)$$

scheme-theoretically. Blow up X along G , obtaining $\tilde{X}$, and let $\tilde{p} : \tilde{Z} \rightarrow \tilde{X}$ be deduced from $p : Z \rightarrow X$ by base change. Then the closure $\overline{Y - p^{-1}(G)}$ in $\tilde{Z}$ of $Y - p^{-1}(G)$, and $\tilde{p}^{-1}(\overline{F - G})$ in $\tilde{X}$, are disjoint closed subsets of $\tilde{Z}$.

We may assume that Z and X are affine. Let $\mathfrak{G}$ be the ideal of G . Cover $\tilde{X}$ by the opens $D(g)$, for g in $\mathfrak{G}$ (a system of generators of $\mathfrak{G}$ suffices), and cover $X - G$ by

$$U(g) = D(g) - (\text{inverse image of } G).$$

Thus

$$\begin{array}{ccc} X - G & \hookrightarrow & \tilde{X} \\ \parallel & & \parallel \\ \bigcup U(g) & \hookrightarrow & \bigcup D(g). \end{array}$$

It is enough to see that, for each g , the traces on $\tilde{p}^{-1}D(g)$ of $\tilde{p}^{-1}(\overline{F-G})$ and of $\overline{Y-p^{-1}G}$ are disjoint. They are respectively the inverse image of the closure of $F \cap U(g)$ in $D(g)$, and the closure of $Y \cap p^{-1}(U(g))$ in $\tilde{p}^{-1}D(g)$.

Let $\sim$ denote an inverse image from X to Z , or from $D(g)$ to $\tilde{p}^{-1}D(g)$. Since $p^{-1}G \supset Y \cap p^{-1}(F)$, the inverse image $\tilde{g}$ of g on Z can be written $\tilde{g} = y + f$, with y zero on Y and f zero on $p^{-1}(F)$. There are functions a_i on Z and elements f_i in the ideal of F such that $f = \sum a_i \tilde{f}_i$. Since $F \supset G$, the f_i lie in the ideal of G and the f_i/g are functions on $D(g)$. They vanish on $F \cap U(g)$, hence on $\overline{F \cap U(g)}$ in $D(g)$, and $\sum a_i (f_i/g)^\sim$ is zero on $\tilde{p}^{-1}(\overline{F \cap U(g)})$. On $p^{-1}U(g)$, one has

$$1 = y/g + f/g = y/g + \sum a_i (f_i/g)^\sim,$$

and the same sum is therefore equal to 1 on $Y \cap p^{-1}U(g)$, hence on its closure in $\tilde{p}^{-1}D(g)$. This proves the required disjointness.

Remark. If one is instead given closed subsets $F \supset G_0$ of X , with G_0 such that $Y \cap p^{-1}(F)$ is set-theoretically contained in $p^{-1}(G_0)$, then there is an r such that the r -th infinitesimal neighbourhood G of G_0 in F satisfies the hypotheses of the lemma. If $\mathfrak{J}$ is the ideal defining $(Y \cap p^{-1}(F))_{\text{red}}$ in $Y \cap p^{-1}(F)$, it is enough to take r such that $\mathfrak{J}^{r+1} = 0$. Thus the lemma makes it possible, by a blow-up which does not touch the complement V of G_0 , to “separate” the proper transform of F from $\overline{Y-p^{-1}(G_0)}$ in $\tilde{Z}$.

Lemma (0.4). *Let there be a commutative diagram of schemes*

$$\begin{array}{ccccc} & & X' & \hookrightarrow & \overline{X}' \\ & \nearrow & \downarrow p & & \downarrow q \\ U & \hookrightarrow & X & \hookrightarrow & \overline{X} \end{array}$$

with U a dense open subset of $\overline{X}$ and $\overline{X}'$, with p and q separated of finite type, and with $q^{-1}(X) = X'$. Let F be a closed subset of X , and let G be a closed subset of $\overline{X}'$ containing $p^{-1}(F)$. Then there exists an ideal $\mathfrak{a}$, with $V(\mathfrak{a}) \subset \overline{F} - F$, such that, after base change by the blow-up morphism defined by $\mathfrak{a} : \tilde{X} \rightarrow \overline{X}$, and after replacing the new $\overline{X}'$ by the closure of U , one has

$$q^{-1}(\overline{F}) \subset G.$$

In Nagata’s language, to take the closure of F in the Zariski–Riemann space of X (“bubble space”), it is enough to take the “intersection” of the closures of F in the blow-ups of X with centre contained in $\overline{F} - F$.

Apply Lemma 0.3 to

$$\begin{array}{ccccc} \overline{X}' - q^*\overline{F} & \hookrightarrow & & \hookrightarrow & \overline{X}' - G \\ \downarrow & & & & \downarrow \\ \overline{X} - \overline{F} & \hookrightarrow & \overline{X} - (\overline{F} - F) & \hookrightarrow & \overline{X}. \end{array}$$

One obtains $\mathfrak{a}$, and, after the base change $\tilde{X} \rightarrow \overline{X}$, the new closure $\overline{F}$ is disjoint from the image of the closure of the old $\overline{X}' - q^*\overline{F}$ in $\overline{X}' - G$; in particular, after replacing $\tilde{X}$ by $\overline{U}$, this new closure is disjoint from the image of $\overline{X}' - G$, as required.

Lemma (0.5). *Let there be a diagram of separated finite type S -schemes*

$$\begin{array}{ccc} & X_i & \hookrightarrow \overline{X}_i \\ & \downarrow & \\ U & \hookrightarrow & X \end{array} \quad (1 \leq i \leq n)$$

with $\overline{X}_i$ proper over S , and U dense open in X and $\overline{X}_i$. Let X^* be the closure of U in the product of the $\overline{X}_i$. Finally, let F_i be a closed subset of X_i ; assume that the intersection of the inverse images of the F_i in X^* is empty. Then, after replacing $\overline{X}_i$ by a blow-up with centre in $\overline{F}_i - F_i$, one can obtain an analogous situation in which the intersection of the inverse images of the $\overline{F}_i$ is empty.

Let X^{**} be a blow-up of X^* , $X^{**} \xrightarrow{p_i} \overline{X}_i$, such that in X^{**} one has

$$\bigcap_i \overline{p_i^{-1}(F_i)} = \emptyset$$

(blow up $\bigcap_i p_i^{-1}(F_i)$). Apply Lemma 0.4 to the diagrams

$$\begin{array}{ccccc} & & p_i^{-1}(X_i) & \hookrightarrow & X^{**} \\ & \nearrow & \downarrow & & \downarrow \\ U & \hookrightarrow & X_i & \hookrightarrow & \bar{X}_i \end{array}$$

and to $F_i \subset X_i$, $G_i = p_i^{-1}(F_i)$. This gives the desired blow-up of the $\bar{X}_i$. Indeed, in the new situation, in the closure of U in X^{**} , one has

$$\bigcap_i p_i^{-1}(\bar{F}_i) \subset \bigcap_i G_i = \emptyset.$$

Since $X^{**} \rightarrow X^*$ is surjective, this remains true in X^* .

1. The theorems

Definition (1.1). Let X and Y be two S -schemes, with Y separated over S . A quasi-domination $f : X \rightarrow Y$ is a pair consisting of a dense open subset U of X and a morphism of S -schemes $f : U \rightarrow Y$ whose graph is closed in $X \times_S Y$.

Let X and Y be two S -schemes, with Y separated over S , let U be a (quasi-compact) open subset of X , and let $f : U \rightarrow Y$. We say that X is quasi-dominant (relative to U and f) over Y if there exists a quasi-domination of the schematic closure $\bar{U}$ of U , with values in Y , extending f . There is at most one such quasi-domination, whose graph is the closure of the graph of f . A quasi-domination f is said to be proper (or complete) if the morphism $f : U \rightarrow Y$ is proper.

Theorem (1.2, [2, Theorem 3.2]). Let X and Y be two S -schemes and let $U \subset V \subset X$ be two dense open subsets of X . Assume that Y is separated and of finite type over S . Let $f : U \rightarrow Y$ be a quasi-domination of V into Y . There exists an ideal $\mathfrak{a}$, with $V(\mathfrak{a}) \subset X - V$, such that the blow-up $\tilde{X}$ of X along $V(\mathfrak{a})$ quasi-dominates Y .

By the base change $X \rightarrow S$, we are reduced to assuming $X = S$. Then Y is a scheme over X , f is a section of Y above U , and, by hypothesis, $\overline{f(U)}$ is closed above V .

Case 1. $U = V$, $p : Y \rightarrow X$ quasi-affine, X affine. By hypothesis there is a commutative diagram

$$\begin{array}{ccc} Y & \xrightarrow{\quad} & E_X \\ & \searrow & \swarrow \\ & X & \end{array}$$

with E_X an affine-space type over X . If $\tilde{X}$ quasi-dominates E_X (relative to f), then a fortiori X quasi-dominates Y . We may therefore assume that $Y = E_X$. A preliminary blow-up allows one to assume that $X - U$ is a divisor D . Locally on X , f then has coordinates $(f_i u^{-n})$, where u is a local equation of D , where f_i is regular, and where n is chosen sufficiently large once and for all. Let $\mathfrak{a}$ be the ideal generated by u^n and the f_i . After blowing up $V(\mathfrak{a})$, f extends to a section of the completed projective space of E_X , and this proves the case.

Case 2. $U = V$, p quasi-affine. By Case 1, if $(U_i)_{0 \leq i \leq n}$ is an affine covering of X , then on U_i there exists a suitable ideal $\mathfrak{a}_i$. This ideal has an extension $\tilde{\mathfrak{a}}_i$ on $\tilde{X}$, with $V(\tilde{\mathfrak{a}}_i) \subset X - V$, and it is enough to blow up the product of the $\tilde{\mathfrak{a}}_i$.

Case 3. p quasi-affine. By Lemma 0.3 applied to

$$\begin{array}{ccc} \overline{f(U)} & \hookrightarrow & Y \\ \downarrow & & \downarrow p \\ U & \hookrightarrow & V \hookrightarrow X \end{array}$$

a preliminary blow-up allows us to assume that $p(\overline{f(U)}) \cap (V - U) = \emptyset$. There is therefore a quasi-compact open X' of X with $X' \cap V = U$ and $p(\overline{f(U)}) \subset X'$. By Case 2, there exists an ideal $\mathfrak{a}$ on X' , with $V(\mathfrak{a}) \subset X' - U$, such that the blow-up $\tilde{X}'$ quasi-dominates Y , that is, $\overline{f(U)}$. It remains to extend $\mathfrak{a}$ to an ideal on X , with $V(\mathfrak{a}) \subset X - V$.

General case. Let (Y_i) be a finite covering of Y by affine opens, and put

$$U_i = f^{-1}(Y_i), \quad X_i = \bar{U}_i, \quad V_i = X_i \cap V.$$

Case 3 applies to

$$\begin{array}{ccccc} & & Y_i \cap p^{-1}(X_i) & & \\ & \nearrow & \downarrow & & \\ U_i & \hookrightarrow & V_i & \hookrightarrow & X_i. \end{array}$$

and gives an ideal $\mathfrak{a}_i$ on X_i ; this ideal defines an ideal $\mathfrak{a}_i$ on X , still with $V(\mathfrak{a}_i) \subset X - V$. Blow up the product of the $\mathfrak{a}_i$. The resulting blow-up $q : \tilde{X} \rightarrow X$ is such that $q^{-1}(X_i)$ dominates the blow-up $\tilde{X}_i$. If $X'_i = q^{-1}(X_i)$, there is therefore an open subset U'_i of X_i , with $U'_i \supset q^{-1}(U_i)$, such that, over X_i , $\overline{f(U_i)}$ is a section of $f : U'_i \rightarrow Y$. It follows that, after blowing up,

$$\overline{f(U)} = \bigcup_i f_i(U'_i).$$

To prove Theorem 1.2, we may replace Y by $\overline{f(U)}$, and conclude by Case 3 and the following lemma.

Lemma (1.3). *$\overline{f(U)}$ is quasi-affine over $\tilde{X}$.*

Indeed, $\overline{f(U)}$ is quasi-finite over $\tilde{X}$.

Corollary (1.4, a variant of Chow's lemma). *Let $f : X \rightarrow S$ be a separated morphism of finite type. Then for every open U , quasi-projective over S and dense in X , there exists a diagram of S -schemes*

$$X \xleftarrow{q} X' \xrightarrow{j} \overline{X}$$

where q is a blow-up morphism defined by an ideal $\mathfrak{a}$ satisfying $V(\mathfrak{a}) \subset X' - U$, and where $\overline{X}$ is projective over S .

In particular, q is projective and surjective and $q^{-1}(U) \simeq U$.

Let U^* be a projective S -scheme containing U as a schematically dense open. By Theorem 1.2 applied to the inclusion of U in X , one can choose U^* quasi-dominant over X : there is a diagram

$$\begin{array}{ccccc} U & \hookrightarrow & V & \hookrightarrow & U^* \\ \parallel & & \downarrow \varphi & & \\ U & \hookrightarrow & X & & \end{array} \quad (1.4.1)$$

and here, since U^* is proper over S , φ is proper.

Lemma (1.5). *Let there be a diagram of separated finite type S -schemes*

$$\begin{array}{ccccc} U & \hookrightarrow & V & \hookrightarrow & Y \\ \parallel & & \downarrow \varphi & & \\ U & \hookrightarrow & X & & \end{array}$$

with U dense in X , dense in Y , and schematically dense in V , and with φ a quasi-domination of Y in X . If $\mathfrak{a}$ is an ideal of X , with $V(\mathfrak{a}) \subset X - U$, such that the blow-up $\tilde{X}$ of X along $\mathfrak{a}$ quasi-dominates V (relative to U), and if U is schematically dense in $\tilde{X}$, whence a diagram

$$\begin{array}{ccccc} U & \hookrightarrow & W & \hookrightarrow & \tilde{X} \\ & \searrow & \downarrow \psi & & \\ & & V & & \end{array}$$

then:

(a) ψ is the blow-up morphism of V along $\varphi^*\mathfrak{a}$;

(b) the graph of ψ is closed in $\tilde{X} \times_S Y$.

Let $\tilde{V}$ be the blow-up of V along $\varphi^*\mathfrak{a}$. By the universal property of $\tilde{X}$, there is a unique X -morphism from

$\tilde{V}$ to $\tilde{X}$, call it α . By the universal property of $\tilde{V}$, there is a unique V -morphism from W to $\tilde{V}$, call it β :

$$\begin{array}{ccccc}
 & & V & \longrightarrow & Y \\
 & \nearrow & \downarrow \varphi & & \\
 U & \longrightarrow & W & \longrightarrow & \tilde{X} \\
 & \searrow & \downarrow \beta & \nearrow \alpha & \\
 & & \tilde{V} & & \\
 & & \downarrow & & \\
 & & X & &
 \end{array}$$

By Theorem 1.2, by blowing up X_1 without touching U , we are reduced to the case in which X_1 quasi-dominates X_2 :

$$\begin{array}{ccc}
 & V & \hookrightarrow X_1 \\
 & \downarrow & \\
 U & \begin{array}{c} \nearrow \\ \searrow \end{array} & X_2
 \end{array}$$

Apply Lemma 1.5 to this diagram, in order to obtain

$$\begin{array}{ccc}
 & W & \hookrightarrow X'_2 \\
 & \downarrow \psi & \\
 U & \begin{array}{c} \nearrow \\ \searrow \end{array} & V \hookrightarrow X_1
 \end{array}$$

where ψ is the blow-up morphism of an ideal $\mathfrak{a}$ of V with $V(\mathfrak{a}) \subset V - U$, the graph of ψ is closed in $X_1 \times X'_2$, and X'_2 is a blow-up of X_2 .

Let $\mathfrak{a}'$ be an ideal of X_1 extending $\mathfrak{a}$, with $V(\mathfrak{a}') \subset X_1 - U$, and let $q : X'_1 \rightarrow X_1$ be the blow-up of X_1 along $\mathfrak{a}$. Then ψ induces an isomorphism between W and $q^{-1}(V)$, and the graph of this isomorphism is closed in $X'_1 \times X'_2$. Define X by gluing X'_1 and X'_2 along this isomorphism.

Remark. The proof shows that one may take $p_i : U_i \rightarrow X_i$ ($U_i \subset X$) to be a blow-up morphism of an ideal away from U .

Theorem (1.6, [2, Lemma 4.1 and Theorem 4.3]). *Let $f : X \rightarrow S$ be separated of finite type. There exists an immersion of X into an S -scheme proper over S .*

The graph of φ is (trivially) proper over V ; the same is true of its inverse image on $V \times_S \tilde{X}$, and it follows that the graph of ψ is proper over V , i.e. that ψ is proper. Thus β is proper, and has schematically dense image, since already U is schematically dense in $\tilde{V}$. On the other hand, β is a local section of α on W (because $\beta|_U$ is one on U), hence β is an isomorphism.

By hypothesis, the graph of ψ is closed in $V \times_S \tilde{X}$, and is contained in the inverse image of the graph of φ , which is closed in $Y \times X$; hence $\Gamma(\psi)$ is closed in $Y \times \tilde{X}$.

Apply Lemma 1.5 to (1.4.1). There exists an ideal $\mathfrak{a}$ on X , with $V(\mathfrak{a}) \subset X - U$, such that the blow-up $\tilde{X}$ quasi-dominates V . One can moreover arrange that U be schematically dense in $\tilde{X}$. By Lemma 1.5, and because φ is proper, $\tilde{X}$ is therefore the blow-up of V along $\varphi^*\mathfrak{a}$. If $\mathfrak{a}'$ is an ideal of U^* extending $\mathfrak{a}$, with $V(\mathfrak{a}') \subset U^* - U$, one then takes

$$X' = \tilde{X}, \quad \bar{X} = \tilde{U}^*.$$

Lemma (1.7, [2, Lemma 4.2]). *Let there be a diagram of separated finite type S -schemes*

$$X_2 \leftarrow U \rightarrow X_1$$

with U a dense open subset of the X_i . There exist $j : U \rightarrow X$, with X separated of finite type over S and U schematically dense in X , and proper quasi-dominations $p_i : X \rightarrow X_i$.

Let U_i be a finite open covering of X by dense quasi-projective opens. For each U_i , there exists, see Corollary 1.4, a commutative diagram of S -schemes

$$\begin{array}{ccc} & X'_i & \xrightarrow{j_i} \overline{X}_i \\ & \nearrow q_i \downarrow & \\ U_i & \longrightarrow & X_i \end{array}$$

with q_i proper, $\overline{X}_i$ proper over S , and U_i dense in $\overline{X}_i$. Put $F_i = \overline{X}_i - U_i$ and $F'_i = q_i^{-1}(F_i)$. Finally, let X^* be the closure of U in the product of the $\overline{X}_i$:

$$p_i : X^* \longrightarrow \overline{X}_i.$$

By Lemma 0.5, a preliminary blow-up of $\overline{X}_i$, not touching X'_i , allows us to assume that

$$\bigcap_i p_i^{-1}(F'_i) = \emptyset.$$

Let M_i be the separated S -scheme obtained by gluing X and $\overline{X}_i - F'_i$ along U_i :

$$\begin{array}{ccc} & \overline{X}_i - F'_i & \\ & \downarrow & \\ X & \hookrightarrow & M_i. \end{array}$$

Apply Lemma 1.7 to X and the M_i , to obtain $X \rightarrow M$, where M properly quasi-dominates each of the M_i . It remains to prove that M is proper over S . Indeed, one has maps

$$X^* - p_i^{-1}(F'_i) \longrightarrow M_i$$

which glue over $U = \bigcap_i U_i$. Replacing X^* by a blow-up X^{**} , one obtains maps

$$X^{**} - p_i^{-1}(F'_i) \longrightarrow M$$

which glue; this gives a map from the schematic closure of U in X^{**} to M . The scheme M , being the image of this map, is proper over S .

Comments

I hope that, by being still more constructive, one could prove Theorem 1.6 for $f : X \rightarrow S$ separated of finite type, with S quasi-compact and quasi-separated. The rules to follow are:

- (a) consider only quasi-compact opens and closed subsets defined by ideals of finite type;
- (b) never take schematic closures (one would leave (a)), but only take a closed subset containing the given part and sufficiently small for the later constructions;
- (c) do not speak of dense quasi-projective opens (cf. (b)). Remember that, to construct such opens, one could start from an affine covering U_i , and take

$$U = \bigcup_i \left(U_i - \bigcup_{j < i} U_j \right).$$

Translate the arguments in terms of the original U_i , and apply (a) + (b).

On the other hand, I have no idea in the case of algebraic spaces.

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